M6.4.2 Multiply Decimals – Student Version

Opening

1. Write 2 words to describe how you are feeling today.

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2. Which one doesn't belong? Be prepared to explain why.

A 0.04	B 0.4
C $\frac{4}{100}$	D $\frac{1}{10} \cdot \frac{4}{10}$

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Background Skills/Do Now

1. Fill in each blank with the value that makes the statement true.

a) $12.3 \cdot$ _____ $= 123$

b) $0.123 \cdot 100 =$ _____

c) $123 \cdot 0.01 =$ _____

2. What is the product of $269 \cdot 43$? Show your work.

3. a) Convert each fraction to a decimal.

i. $\frac{4}{10}$

ii. $\frac{8}{100}$

b) Convert each decimal to a fraction.

i. 0.9

ii. 0.11

4. *Spiral Review or Review from Yesterday*

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Think!

Evelyn applied her understanding of fractions to find the product of 0.2 and 0.3. Her work is shown below.

$$(0.2) \cdot (0.3) = \frac{2}{10} \cdot \frac{3}{10} = \frac{6}{100} = 0.06$$

1. Explain the steps Evelyn took to find the product of 0.06.

2. Use Evelyn's method to calculate each product below.

a) $(0.2) \cdot (0.03)$

b) $(0.02) \cdot (0.03)$

3. Look carefully at the number of decimal places in the factors and the number of decimal places in the product in all three multiplication problems. Describe the pattern you notice.

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4. Use your conclusion from the previous problem to correctly place the decimal point in each product below, given that $96 \cdot 437 = 41{,}952$.

a) $(0.96) \cdot (43.7) = 4\ 1\ 9\ 5\ 2$

b) $(9.6) \cdot (0.437) = 4\ 1\ 9\ 5\ 2$

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Examples

Example 1

Multiply.

$$(1.2) \cdot (1.1)$$

Example 2

Multiply.

a) $(2.95) \cdot (4.1)$

b) $(0.51) \cdot (0.032)$

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Practice Problems

Level 1

1. a) The correct digits of each product are given below. Place the decimal in the correct place in each product.

i. (87.49) · 32 = 2 7 9 9 6 8

ii. (13.42) · (21.75) = 2 9 1 8 8 5 0

iii. (24.85) · (19.3) = 4 7 9 6 0 5

b) Choose one of the problems above and explain how you know you placed the decimal correctly in the product.

2. Fill in each blank using the word bank below.

sum	product	factor
quotient	one	tenth
hundredth	two	decimal

In the multiplication problem (5.9) · (7.4) = 43.66, 5.9 is a _____ and 43.66 is the _____.

Multiplying by one _____ moves the decimal point one place to the left.

Multiplying by one hundredth moves the decimal point _____ places to the left.

3. Which equation is true?

A. (219.8) · (4.3) = 9.4514

B. (21.98 · (0.43) = 945.14

C. (0.2198) · (43) = 945.14

D. (21.98) · (4.3) = 94.514

4. Which expression doesn't belong? Explain your reasoning.

A. 4 · (0.9)

B. 4 · 9 · (0.1)

C. 36 · $\frac{1}{10}$

D. 3.6

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5. Find each product. Show your work.

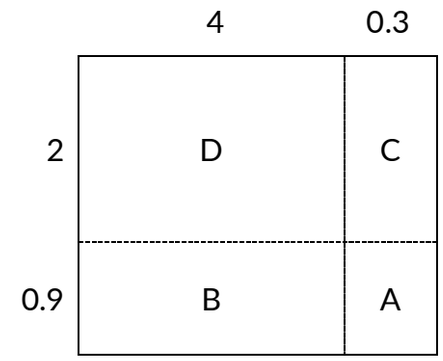
a) (2.43) · (4.8)

b) (0.053) · (92.7)

c) (5.28) · (0.31)

6. A rectangle has been partitioned into four smaller rectangles.

a) Write the letter of the smaller rectangle area that can be represented by each numerical expression.



i. 2 · (0.3)

ii. 2 · 4

iii. 4 · (0.9)

iv. (0.3) · (0.9)

b) How could we use the area of the four smaller rectangles to find the area of the big rectangle?

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7. Use the product of 5 and 7 to find (0.05) · (0.007).

8. Eryn ran 3.72 miles during track practice. Kaylah ran 2.5 times the distance that Eryn ran. How many miles did Kaylah run?

9. Find each product below, given that 39 · 23 = 897.

a) (3.9) · (2.3)

b) 0.039 · 23

c) (0.39) · (0.23)

10. Match each expression on the left with its product.

Column A	Column B
a) (32.5) · (2.41)	i. 0.78325
b) (0.325) · (24.1)	ii. 783.25
c) (32.5) · (24.1)	iii. 78.325
d) (3.25) · (0.241)	iv. 7.8325

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11. Jayden used place value and properties of operations to find the product of (2.1) · (0.03). He accidentally left some parts of his work blank.

$(2.1) \cdot (0.03) =$

$21 \cdot (0.1) \cdot 3 \cdot \square =$

$63 \cdot \square =$

0.063

a) What number must go in the first box for his work to be correct?

b) What number must go in the second box for his work to be correct?

12. A cafeteria table is 12.04 feet long and 5.9 feet wide.

a) Use estimation to determine the area of the table without calculating it exactly.

$$(12.04) \cdot (5.9)$$

A. 7,103.6 feet squared

B. 710.36 feet squared

C. 71.036 feet squared

D. 7.1036 feet squared

b) Explain your estimation strategy.

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13. a) Andrea tried to used fraction multiplication to help her find the product of (1.1) · (0.9). Her work is shown below. In what step did she make her first error?

Step 1	$\frac{11}{100} \cdot \frac{9}{10} =$
Step 2	$\frac{99}{1{,}000} =$
Step 3	0.099

A. Step 1

B. Step 2

C. Step 3

D. There is no error.

b) Explain Andrea's error.

14. a) For each problem, use estimation to determine whether the product will be greater than 1 or less than 1.

i. (1.2) · (2.3)

ii. (0.9) · (0.85)

iii. (1.05) · (3.09)

iv. (0.875) · (0.625)

b) Explain your answer to part iv.

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Level 2

15. Abigail tried to find the product of 2.9 and 7.3. Her work and explanation are shown below.

$$2.9 \cdot 10 = 29$$
$$7.3 \cdot 10 = 73$$
$$29 \cdot 73 = 2{,}117$$
$$2{,}117 \div 10 = 211.7$$

Because I multiplied each factor by 10 to make them into whole numbers, I divided the product of 29 and 71 by 10 to get my answer.

a) Use estimation to explain why Abigail's answer is not reasonable.

b) Help Abigail to understand and correct her error.

16. Multiply.

a) $(0.46) \cdot (83.512)$

b) $(0.048) \cdot (0.39)$

c) $(12.4) \cdot (4.35)$

17. a) Which problems will have the same product as $(4.73) \cdot (8.61)$? Select each correct answer.

A. $(47.3) \cdot (0.861)$

B. $(0.473) \cdot (0.861)$

C. $(47.3) \cdot (86.1)$

D. $(0.473) \cdot (86.1)$

E. $(0.473) \cdot 861$

b) Explain how you selected one of your answer choices.

18. Malik tried to use partial products to find the product of 6.4 and 3.7. His work is shown below. Help Malik to understand and correct his error.

$$(6.4) \cdot (3.7)$$
$$6 \cdot 3 = 18$$
$$(0.4) \cdot (0.7) = 0.28$$
$$18 + 0.28 = 18.28$$

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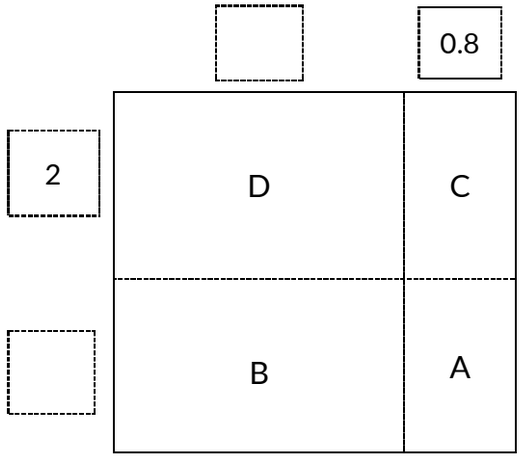
19. The area model below will be used to find the product of (3.8) · (2.6).

a) Label the missing side lengths of the area model.



b) Find the area of each of the smaller rectangles in square units.

c) Use the areas of the smaller rectangles to find the product of (3.8) · (2.6).

20. Ricardo tried to find the product of 0.04 and 1.456. His work is below.

```
  1 2 2
  1.4 5 6
× 0.0 4
  5.8 2 4
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Identify and correct the error Ricardo made.

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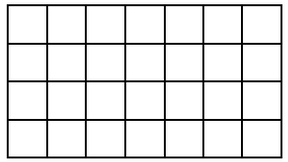
21. The area model below represents the product of 0.4 and 0.7.



a) What is the side length of each square in the model? Explain how you know.

b) What is the area of each square in the model?

c) What is (0.4) · (0.7)?

22. At the local market, tomatoes are on sale for $1.25 per pound. Which of the following expressions correctly represents the cost of 0.6 pounds of tomatoes?

A. (1.25) · (0.6) = 0.75

B. (125) · (6) = 75

C. (1.25) · (0.6) = 7.5

D. (1.25) · (0.6) = 0.075

23. Which expressions have the same value as (8.2) · (0.57)? Select each correct answer.

A. 82 · (0.1) · 57 · (0.1)

B. 82 · (0.1) · 57 · (0.01)

C. 82 · (0.01) · 57 · (0.01)

D. $82 \cdot \frac{1}{10} \cdot 57 \cdot \frac{1}{10}$

E. $82 \cdot \frac{1}{10} \cdot 57 \cdot \frac{1}{100}$

24. Diamond is deciding where to place the decimal after multiplying 2.48 by 7.16. She said, “There are two decimal places in each of the factors, so there will be two decimal places in the product.” Do you agree? Explain your reasoning.

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25. At a grocery store, blueberries cost $1.59 per pound. Kimora bought 2.5 pounds of blueberries. How much did she pay? Round your answer to the nearest cent.

26. Gasoline costs $2.19 per gallon. Neveah filled her car with 11.5 gallons of gas. How much did she spend? Round your answer to the nearest cent.

27. A rectangular field has a length of 1.25 km and a width of 0.75 km. What is the area of the field?

28. a) How many times greater will the product of $396 \cdot 429$ be than the product of $3.96 \cdot (4.29)$?

A. 10 times greater

B. 100 times greater

C. 1,000 times greater

D. 10,000 times greater

b) Explain your reasoning for your answer in part a).

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Level 3

29. Which expressions have the same value as $(0.09) \cdot (0.287)$? Select each correct answer.

A. $9 \cdot (0.01) \cdot 287 \cdot (0.001)$

B. $9 \cdot 287 \cdot (0.00001)$

C. $9 \cdot (0.1) \cdot 287 \cdot (0.01)$

D. $9 \cdot \frac{1}{10} \cdot 287 \cdot \frac{1}{100}$

E. $9 \cdot 287 \cdot \frac{1}{100{,}000}$

30. Write three multiplication expressions that have the same value as $(0.03) \cdot (0.6)$.

31. Find each product. Express each product as a decimal.

a) $(4.3) \cdot 2\frac{1}{2}$

b) $3\frac{4}{5} \cdot (0.62)$

32. Zaniya multiplied two decimals to get the correct product, but she forgot to put the decimal points in her factors. Her work is shown below. What could have been the correct factors? Explain your reasoning.

$$(396) \cdot (472) = 18.6912$$

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33. Use mental math and place value to find the value of n that makes each equation true.

a) $(0.4) \cdot n = 0.024$

b) $(0.07) \cdot n = 5.6$

c) $(0.012) \cdot n = 0.0048$

d) $25 \cdot n = 0.075$

34. a) Create your own real-world problem that could be solved by finding the product of 3.25 and 0.95.

b) Find the product of 3.25 and 0.95.

c) Explain what this product means in the context of your problem.

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35. Rico and Tyler used different methods to find the product of 1.9 and 0.46.

Rico	Tyler
$(1.9) \cdot 10 = 19$	$1.9 = \frac{19}{10}$
$(0.46) \cdot 100 = 46$	$0.46 = \frac{46}{100}$
$19 \cdot 46 = 874$	$\frac{19}{10} \cdot \frac{46}{100} = \frac{874}{1000}$
$874 \div 1{,}000 = 0.874$	$\frac{874}{1000} = 0.874$

a) Explain each student's method.

b) Use Rico's method to find the product of 2.8 and 6.7.

c) Use Tyler's method to find the product of 2.8 and 6.7.

36. Fill in each blank with always, sometimes, or never. Explain your reasoning.

a) A decimal greater than one multiplied by a decimal less than one _____ has a product greater than 1.

b) The sum of the number of decimal places in the factors is _____ equal to the number of decimal places in the product.

c) If the decimal in one factor is moved one place to the left and the decimal in the other factor is moved one place right, the product will _____change.

37. Armando plans to cover his living room with a carpet. His living room is a rectangle that is 14.8 ft by 9.2 ft. The carpet costs $3.50 per square foot. How much will it cost to cover his living room with a carpet?

38. Kendall is putting a wood fence around the edges of her square garden. The side length of the garden is 8.6 feet, and wood fences cost $18.95 per linear foot. How much will she spend?

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39. Apples cost $1.75 per pound and bananas cost $0.20 each. Jamya wants to buy 3.5 pounds of apples and 6 bananas. How much will she spend? Round your answer to the nearest cent.

40. A recipe for one loaf of bread calls for 4.25 cups of flour. Abigail wants to make $6\frac{1}{2}$ loaves of bread. How much flour will she need? Round your answer to the nearest tenth of a cup.

41. Marco is asked to use mental math to find 80 · (0.25). He made the statement below.

"The product is 20 because 0.25 is equivalent to $\frac{1}{4}$ and one fourth of 80 is 20."

Use Marco's strategy to help find the following products. Explain your reasoning for each.

a) 80 · (2.5)

b) 80 · (1.25)

c) (0.8) · (1.25)

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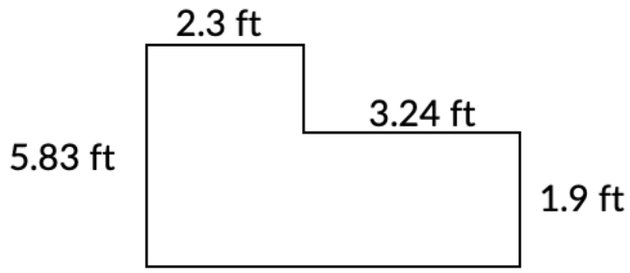
Challenge

42. All of the angles in the hexagon below are right angles. What is the area of the hexagon? Show your work.



43. A rectangular banner is 30 inches long and 17.5 inches wide. There are 2.54 centimeters in an inch.

What is the area of the banner, in square centimeters?

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Closing

1. What method below do you prefer to use to multiply decimals?

Area model:

	2	0.5
1	2	0.5
0.3	0.6	0.15

$$\begin{array}{r} 2.5 \\ \times\ 1.3 \\ \hline 0^{1}.15 \\ 0.6 \\ 0.5 \\ +\ 2.0 \\ \hline 3.25 \end{array}$$

Convert to fractions:

$$2\frac{5}{10} \cdot 1\frac{3}{10} = \frac{25}{10} \cdot \frac{13}{10}$$

$$= \frac{325}{10} = 3.25$$

Decimals:

$$25 \cdot (0.1) \cdot 13 \cdot (0.1)$$

$$= 25 \cdot 13 \cdot (0.1) \cdot (0.1)$$

$$= 325 \cdot (0.01) = 3.25$$

The method I prefer is ___ because...

2. Suppose your friend is not sure where to put the decimal point in the product, shown below.

$$(4.13) \cdot (2.1)$$

$$\begin{array}{r} 413 \\ \times\ \ 21 \\ \hline 413 \\ +\ 8260 \\ \hline 8673 \end{array}$$

What would you tell your friend to help them understand where the decimal point should go?

I would tell them...

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Ticket To Leave

1. What is the product of (12.04) · (5.9)? Show your work.

2. a) Find each product below, given that 248 · 596 = 147,808.

   i. (2.48) · (5.96)

   ii. (24.8) · (59.6)

   b) Explain how you placed the decimal in one of the products in part a).

3. *Fellow's Choice*

4. *Fellow's Choice*

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Ticket to Leave Reflection

1. What were the main concepts, ideas, or skills you learned or improved upon during tutorial?	2. Choose at least one of the following questions to answer: a) What did you learn from a mistake or misconception (either yours or someone else's)? b) How do the concepts you learned today connect to another concept you know?	3. What question(s) do you have about today's work?

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